From orbit to exposure — technical workflow

Satellite mapping of gold-mining ponds and methylmercury capacity, Madre de Dios, Peru, 2016–2025 · companion to poster PRD-PDRDF-012 · CL#26-4012

1 · DATA SOURCES

RAW IMAGERY (THIS STUDY)

- Landsat 7 ETM+, 8/9 OLI (Operational Land Imager) surface reflectance, 30 m — USGS; 10 dry seasons 2016–25
- PlanetScope (PS), 3 m — NASA CSDA (Commercial Smallsat Data Acquisition)
- Sentinel-2 MSI, 10 m — ESA Copernicus

DERIVED BY OTHER GROUPS

- Radar for Detecting Deforestation (RADD) alerts, 10 m — from Sentinel-1 SAR (Synthetic Aperture Radar)
- Visible Infrared Imaging Radiometer Suite (VIIRS) nighttime lights — NOAA
- MERIT Hydro river network
- Amazon Mining Watch (AMW) scars

ECONOMIC + FIELD RECORDS

- Gold price · fish/hair Hg literature

feed

2 · SOURCE ROLES

RAW IMAGERY (THIS STUDY)

- Landsat → pond mapping backbone: only sensor consistent over all 10 yr
- PS → 3 m sub-pixel truth: area calibration + omission QA (30 scenes)
- Sentinel-2 → 10 m check + 2021 fill

DERIVED BY OTHER GROUPS

- RADD → monthly clearing activity, independent of standing water
- VIIRS → monthly brightness of mining camps and towns
- MERIT → flow routing, river prior
- AMW → independent footprint check

ECONOMIC + FIELD RECORDS

- Gold price → economic driver series
- Fish/hair Hg literature → methylation and exposure rates in the findings

classify

3 · SATELLITE MAPPING

- Google Earth Engine (GEE) pipeline
- Water test: Modified Normalized Difference Water Index, MNDWI = (green − SWIR1) / (green + SWIR1), on every clear view of each pixel
- Screens: cloud, shadow, steep slopes, and Landsat 7 Scan Line Corrector (SLC) failure gaps
- Annual rule: a pixel is water that year if ≥50% of its ≥3 clear dry-season views pass the test
- Pond yellowness (red/green) index: active >0.381 · settling <0.209
- Output: annual 30 m class stack (water · pond · bare), 10 bands

mask

4 · TRAINING

- 1,066 hand-labeled river/pond points train and test the river classifiers
- Random forest (RF): an ensemble of decision trees; each tree votes river or pond for a water body and the majority decides
- Oxbow lakes removed by a second RF (70 labels); bare ground by an RF of mining vs natural (594 labels), replacing the point-bar rule (accuracy 0.81 vs 0.47)
- River mask rebuilt for each year (channels migrate); test acc. 0.91
- Ponds = standing water − river mask

verify

5 · VALIDATION

- 2,245 blind hand-labeled points: 585 water · 1,066 river/pond · 594 bare ground
- PlanetScope check: 3 m water maps of 27 scenes: Landsat ~15% low (×1.15, 0.99–1.27; 8 pond-rich footprints), no 2017–25 trend; totals kept uncorrected, so they are conservative
- Ponds missed outright are small (<0.5 ha): ~7% of area but 22% of shoreline; trends unaffected
- 98% of newly detected ponds fall in independently mapped AMW scars
- Sentinel-2 10 m tuned rule: held-out recall 0.77 vs 0.65 (Landsat); cloudy 2021 filled by L+S2, cal. = 7,573 ha

aggregate

6 · OUR DERIVED PRODUCTS

- Annual pond stock + formation-year cohorts, 5 mining zones, 2016–2025
- Active / transitional / settling split from the yellowness index
- Pond area inside vs outside the legal mining corridor set by Legislative Decree 1100 (D.L. 1100)
- Monthly RADD clearing + VIIRS brightness time series
- Pond area routed down the MERIT network to 4 downstream communities
- MeHg capacity (calculated — no new water sampling): mapped pond ha × 5.5, the pond vs river %MeHg ratio measured in MdD waters by Gerson et al. 2020 (n = 12 each, 95% CI 2.9–10.9)

interpret

7 · FINDINGS

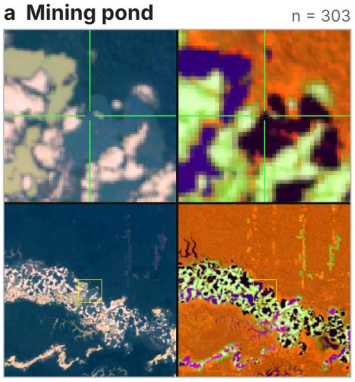
- Pond area doubled (×2.0) from the 2019–20 enforcement baseline (6,517 ha) to 13,061 ha by 2025 — a record high
- Enforcement + COVID cut activity (clearing −44%, worked ponds −57%) but nobody drains a pond: the standing stock never shrank
- Legal corridor: inside grew ×2.22, outside ×1.73 — enforcement relocated mining more than it reduced it
- The pond stock added since 2019 methylates like ≈35,900 ha of new river surface (pond ha × 5.5 ratio, Gerson et al. 2020)
- Upstream pond load: 3,390 ha above Boca Colorado · ~10,550 ha above Laberinto · all 13,061 ha above Puerto Maldonado
- Downstream residents already carry ~2.5× the hair mercury of their neighbors

Validation — the hand labels, and how the map scores

2,245 blind hand labels · final 30-m Landsat classification · companion to poster PRD-PDRDF-012 · CL#26-4012

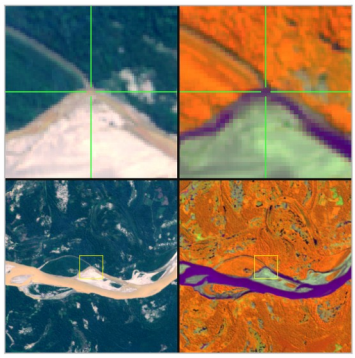
8 · LABEL CLASSES — examples from the labeling games

a Mining pond n = 303



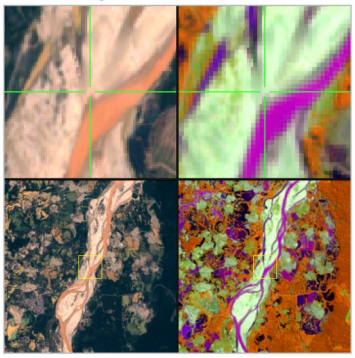
Standing water in an excavated pit among tailings; not connected to the channel network at the ±3 km view.

b River channel n = 316



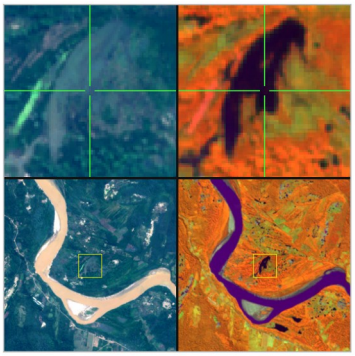
Water continuous with the channel network, including sediment-laden and migrated reaches.

c Wet or point bar n = 151



Unvegetated natural sediment inside the channel belt or on inner bends, wet or dry; not tailings.

d Oxbow lake n = 32



Curved, cut-off meander lake on the floodplain, separate from the active channel.

e Mining bare ground n = 106



Excavated or tailings surface, unvegetated, outside the channel belt.

f Vegetation n = 164



Forest or regrowth canopy at the point, even beside clearings.

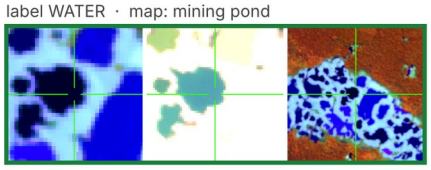
a–d: river–pond game chips (top: true and false colour at ±400 m; bottom: ±3 km context). e–f: bare-ground game chips (true colour). Crosshairs mark the labeled 30-m point. Yearly 10-m Sentinel-2 composites, Madre de Dios, Peru.

9 · MAP ACCURACY — how well the final map predicts standing water

Final map vs 585 hand-labeled water/land points (2019, 2022, 2025): 92.8% of 499 decided points correct · water recall 90% · precision 98%

a Dark standing pond ✓ correct

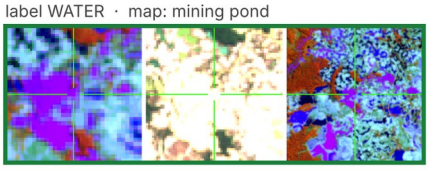
label WATER · map: mining pond



243 water points mapped as pond

b Turbid pond ✓ correct

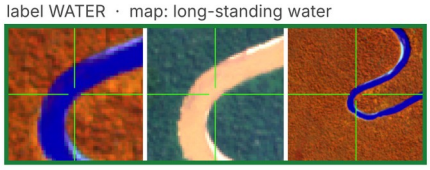
label WATER · map: mining pond



Muddy ponds kept by the permissive water rule

c River channel ✓ correct

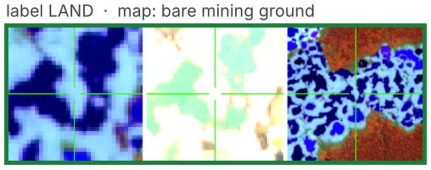
label WATER · map: long-standing water



38 water points mapped as long-standing water

d Tailings between ponds ✓ correct

label LAND · map: bare mining ground



182 of 188 land points correctly not water

e Missed: channel-edge water × error

label WATER · map: not water



30 of 311 water points missed (20 mapped as bare)

f False water: channel bar × error

label LAND · map: long-standing water



Only 6 of 188 land points mapped as water

Each chip, as shown in the labeling game: false colour (water dark blue to purple; tailings pale cyan) · true colour · wider false-colour context; crosshair = labeled 30-m point. Map = final 30-m Landsat classification for that year. 86 water/land-edge points labeled 'unsure' are excluded.